

**A project reports
On**

SMOKE DETECTOR

Submitted by

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Sec -1 Batch – 18

**II/IV BACHELOR OF TECHNOLOGY
IN**



**Department of ELECTRICAL AND ELECTRONICS
ENGINEERING**

K L UNIVERSITY: VADDESWAREM 522502

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K L UNIVERSITY

**DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING**

Analog Electronics Circuit Design (18 EC 2103)

CERTIFICATE

This is to certify that Mr. **M. BHARATH KUMAR – 180069020**, Mr. **M.SAI KUMAR – 180069021**, Mr. **P. HARI – 180069023** of section 1 studying **II/ IV B.Tech in EEE** has satisfactorily completed project **SMOKE DETECTOR** in the semester IV during the academic year **2019 – 2020**

COURSE INSTRUCTOR

COURSE COORDINATOR

SIGNATURE OF HOD

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ABSTRACT

Approximation methods exist to provide estimates of smoke detector response based on optical density, temperature rise, and gas velocity thresholds. The objective of this study was to assess the uncertainty associated with three estimation methods. Experimental data was used to evaluate recommended alarm thresholds and to quantify the associated error. With few exceptions less than 50 percent of the predicted alarm times occurred within 60 seconds of the experimental alarms. At best errors of 20 to 60 percent (in underprediction) occurred for soldering fires using an optical density threshold. For flaming fires, errors in predicted alarm times on the order of 100 to 1000 percent in over prediction of the experimental alarms were common. Overall, none of the approximation methods distinguished themselves as vastly superior. Great care must be exercised when applying these approximation methods to ensure that uncertainty in the predicted alarm times is appropriately considered.

INTRODUCTION ABOUT THIS PROJECT

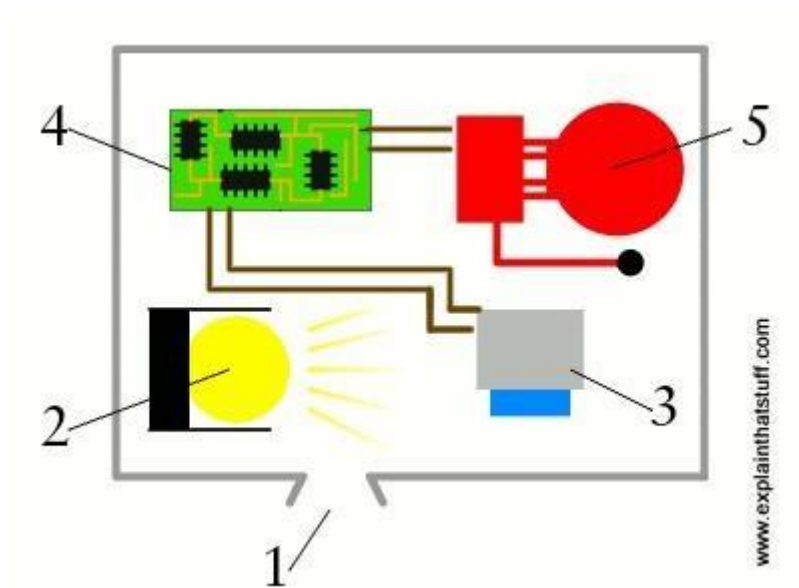
A **smoke detector** is a device that sense smoke, typically as an indicator of fire. Commercial security devices issue a signal to a fire alarm control panel as a part of a fire alarm system, while household smoke detectors, also known as **smoke alarms**, generally issue a local audible or visual alarm from the detector itself or several detectors if there are multiple smoke detectors interlinked.

Smoke detectors are housed in plastic enclosures, smoke detectors are typically shaped like a disk or square about 150 millimeters (6 in) in diameter and 25 millimeters (1 in) thick, but shape and size vary. Smoke can be detected either optically (photoelectric) or by physical process (ionization): detectors may use either, or both, methods. Sensitive alarms can be used to detect, and thus deter, smoking in areas where it is banned. Smoke detectors in large commercial, industrial, and residential buildings are usually powered by a central fire alarm system, which is powered by the building power with a battery backup. Domestic smoke detectors range from individual battery-powered units, to several interlinked mains-powered units with battery backup; with these interlinked units, if any unit detects smoke, all trigger even if household power has gone out.

PRINCIPLE:

The detector must be screwed to your ceiling because that's where smoke heads for when something starts to burn. Fire generates hot gases and because these are less dense (thinner—or weigh less per unit of volume) than ordinary air they rise upward, swirling tiny smoke particles up too. As you can see in the photo up above, the detector has slits around its case (1), which lead to the main detection chamber. An invisible, infrared light beam, similar to the ones that Tom Cruise dodged, shoots into the chamber from a **light-emitting diode (LED)** (2). The same chamber contains a photocell (3), which is an electronic light detector that generates electricity when light falls on it. Normally, when there is no smoke about, the light beam from the LED does not reach the detector. An electronic circuit (4), monitoring the photocell, detects that all is well, and nothing happens. The alarm (5) remains silent.

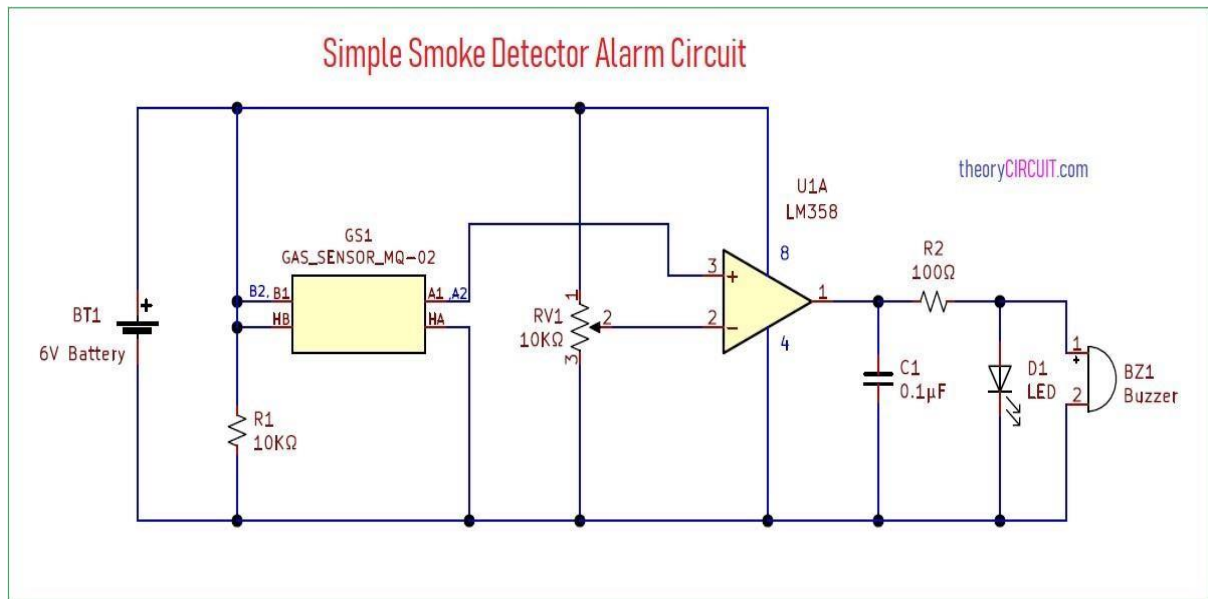
But if a fire breaks out, smoke enters the chamber (6) and scatters some of the light beam (7) into the photocell (3). This triggers the circuit (8), setting off the shrill and nasty alarm (9) that wakes you up and saves your life



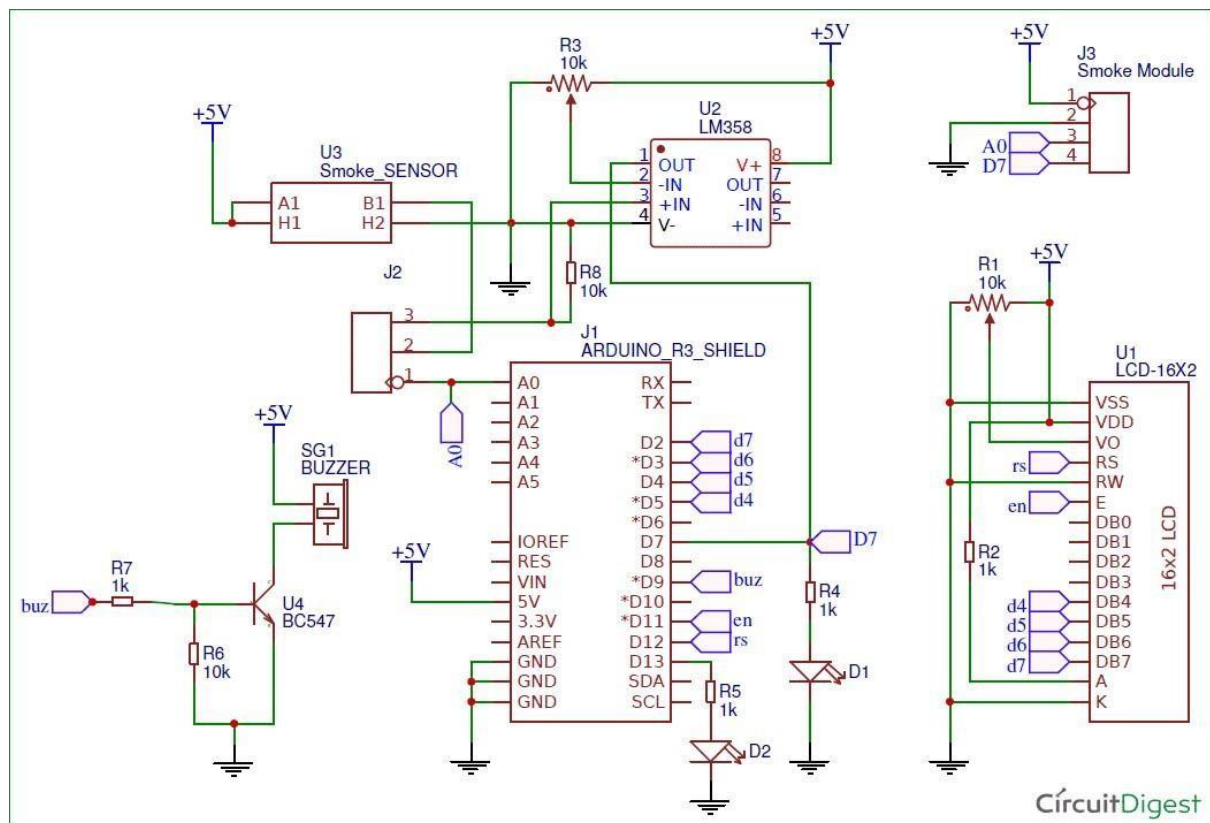
BLOCK DIAGRAM:

Simple Smoke Detector Alarm circuit using MQ02 designed with few easily available components. Main part of this circuit is Gas sensor MQ-02 which is capable of detecting Smoke, LPG, Propane and Hydrogen. It can be used in different types of applications where the smoke and other mentioned gas leak detection required.

The following prototype is a test circuit and it can be constructed through breakout board and PCB after calibration of each components. This smoke detector circuit will produce visible and audible alert when smoke detected.



CONSTRUCTION :-



In this Smoke Detector Circuit with Arduino, we have used a **MQ2 Gas Sensor** to detect preset smoke in the air. A **16x2 LCD** is used for displaying the **PPM value of Smoke**. And an **LM358** IC for converting smoke sensor output into digital form (this function is optional). A **buzzer** is placed as an alarm which gets triggered when smoke level goes beyond 1000 PPM.

Circuit connections for this project are very simple, we have a **Comparator Circuit** for comparing output voltage of smoke sensor with preset voltage (output connected at pin D7). Also smoke sensor output is connected at an analog pin of **Arduino** (A0). Buzzer is connected at Pin D9. And LCD connections are same as Arduino LCD examples that are available in Arduino IDE (12, 11, 5, 4, 3, 2). Remaining connections are shown in the circuit diagram.

COMPONENTS REQUIRED: -

***NODEMCU**

***Bread Board**

***MQ-2 Smoke Sensor**

***Jumper Wires**

***Red Led**

***Green Led**

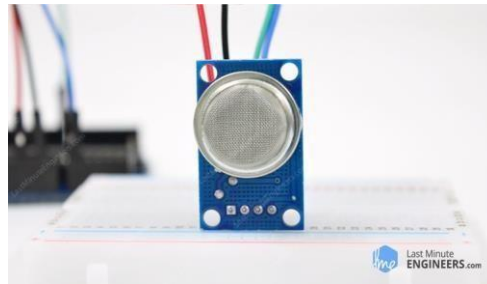
***Resistor**

NODEMCU:

NodeMCU is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online.

In this crystal ball project arduino is used to display the message on the LCD according to the code written in the arduino software, Whenever the tilt switch senses the moment of bread board and sends the signals to the arduino.

MQ-2 SMOKE SENSOR: -



Give your next Arduino project a nose for gasses with the MQ2 Gas Sensor Module. This is a robust Gas sensor suitable for sensing LPG, Smoke, Alcohol, Propane, Hydrogen, Methane and Carbon Monoxide concentrations in the air. If you are planning on creating an indoor air quality monitoring system; breath checker or early fire detection system, MQ2 Gas Sensor Module is a great choice.

MQ2 is one of the commonly used gas sensors in MQ sensor series. It is a Metal Oxide Semiconductor (MOS) type Gas Sensor also known as Chemiresistors as the detection is based upon change of resistance of the sensing material when the Gas comes in contact with the material. Using a simple voltage divider network, concentrations of gas can be detected.



MQ-2 SENSOR

MQ2 Gas sensor works on 5V DC and draws around 800mW. It can detect LPG, Smoke, Alcohol, Propane, Hydrogen, Methane and Carbon Monoxide concentrations anywhere from 200 to 10000ppm.

Here are the complete specifications

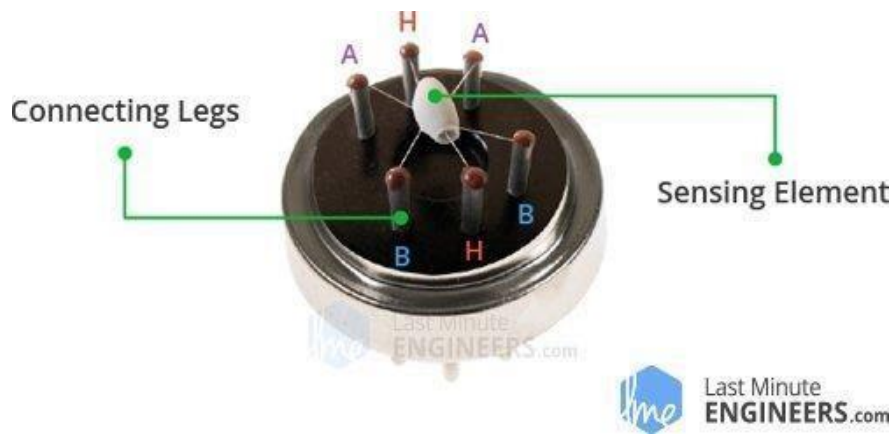
Operating Voltage	5v
Load Resistance	20 K Ω
Heater Resistance	33 $\Omega \pm 5\%$
Heating Consumption	<800mw
Sensing Resistance	10 K Ω – 60 K Ω
Concentration Scope	200 – 10000ppm
Preheat Time	Over 24 hours

Internal structure of MQ2 Gas Sensor

The sensor is actually enclosed in two layers of fine stainless steel mesh called Anti-explosion network. It ensures that heater element inside the sensor will not cause an explosion, as we are sensing flammable gases.



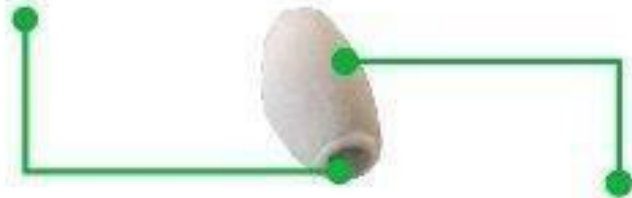
It also provides protection for the sensor and filters out suspended particles so that only gaseous elements are able to pass inside the chamber. The mesh is bound to rest of the body via a copper plated clamping ring.



This is how the sensor looks like when outer mesh is removed. The star-shaped structure is formed by the sensing element and six connecting legs that extend beyond the Bakelite base. Out of six, two leads (H) are responsible for heating the sensing element and are connected through Nickel-Chromium coil, well known conductive alloy.

The remaining four leads (A & B) responsible for output signals are connected using Platinum Wires. These wires are connected to the body of the sensing element and convey small changes in the current that passes through the sensing element.

Aluminum Oxide
(Al₂O₃) based Ceramic

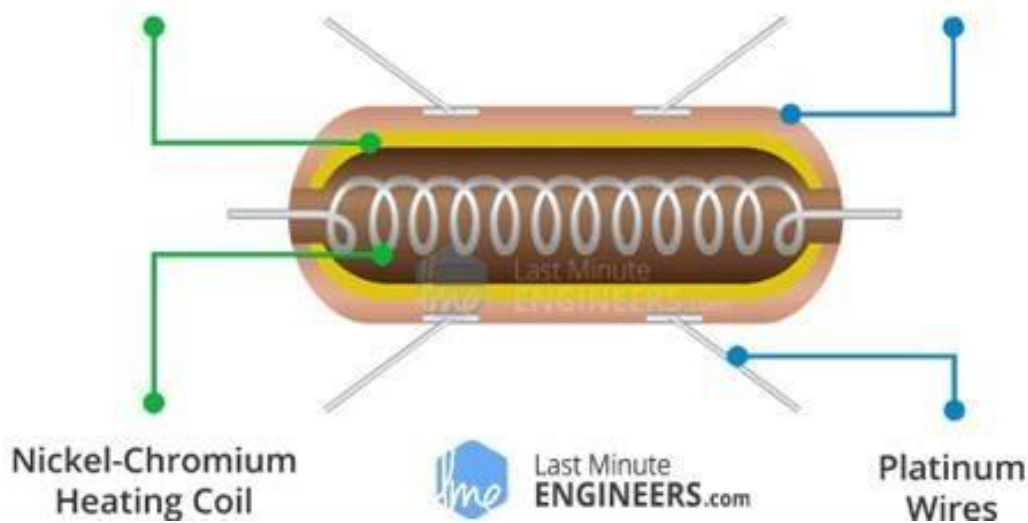


Coating of
Tin Dioxide

The tubular sensing element is made up of Aluminum Oxide (Al₂O₃) based ceramic and has a coating of Tin Dioxide (SnO₂). The Tin Dioxide is the most important material being sensitive towards combustible gases. However, the ceramic substrate merely increases heating efficiency and ensures the sensor area is heated to a working temperature constantly.

Aluminum Oxide
(Al₂O₃) Based Ceramic

Coating of
Tin Dioxide (SnO₂)



So, the Nickel-Chromium coil and Aluminum Oxide based ceramic forms a Heating System; while Platinum wires and coating of Tin Dioxide forms a Sensing System.

How does a gas sensor work?

When tin dioxide (semiconductor particles) is heated in air at high temperature, oxygen is adsorbed on the surface. In clean air, donor electrons in tin dioxide are attracted toward oxygen which is adsorbed on the surface of the sensing material. This prevents electric current flow.

In the presence of reducing gases, the surface density of adsorbed oxygen decreases as it reacts with the reducing gases. Electrons are then released into the tin dioxide, allowing current to flow freely through the sensor

Hardware Overview – MQ2 Gas Sensor

Module

Since MQ2 Gas Sensor is not breadboard compatible, we do recommend this handy little breakout board. It's very easy to use and comes with two different outputs. It not only provides a binary indication of the presence of combustible gases but also an analog representation of their concentration in air.



The analog output voltage provided by the sensor changes in proportional to the concentration of smoke/gas. The greater the gas concentration, the higher is the output voltage; while lesser gas concentration results in low output voltage. The following animation illustrates the relationship between gas concentration and output voltage





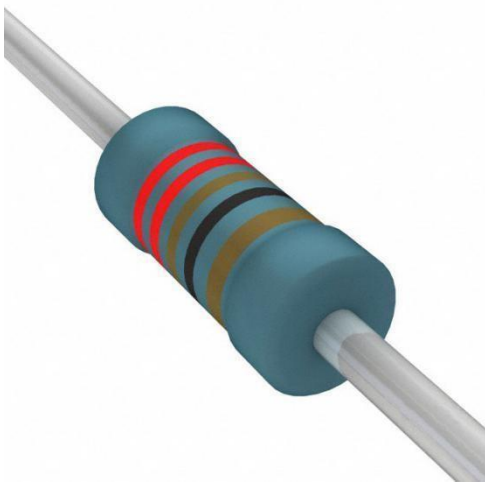
VCC supplies power for the module. You can connect it to 5V output from your Arduino.

GND is the Ground Pin and needs to be connected to GND pin on the Arduino.

D0 provides a digital representation of the presence of combustible gases.

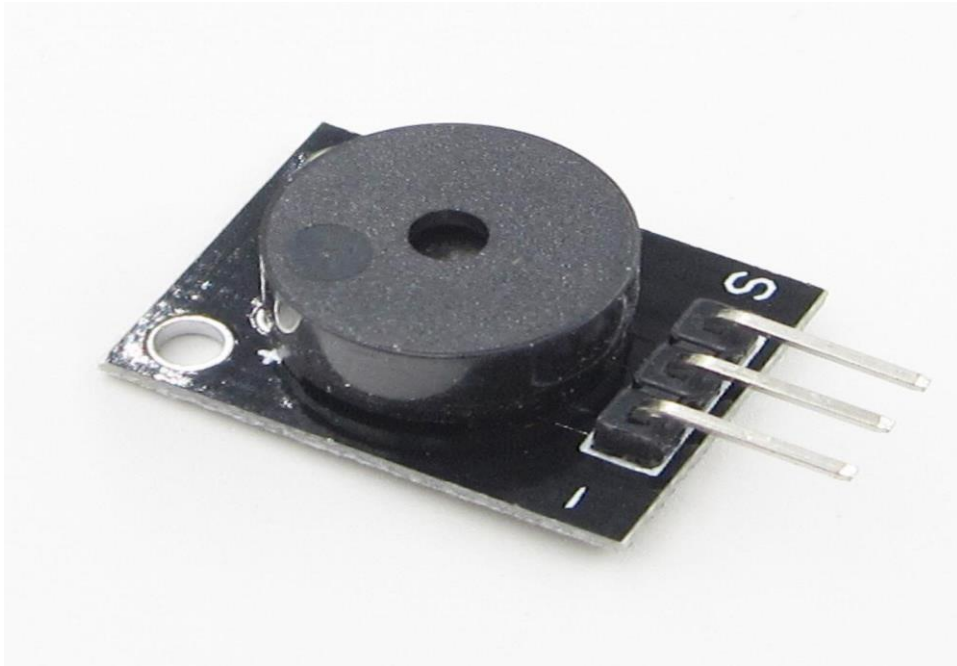
A0 provides analog output voltage in proportional to the concentration of smoke/gas.

RESISTOR:



PASSIVE BUZZER:

A buzzer or beeper is an audio signalling device, which may be mechanical, electromechanical, or piezoelectric (piezo for short). Typical uses of buzzers and Beepers include alarm devices, timers, and confirmation of user input.



JUMPER WIRES:

The term "jumper wire" simply refers to a conducting wire that establishes an electrical connection between two points in a circuit. You can use jumper wires to modify a circuit or to diagnose problems in a circuit. The following steps outline how you can safely use jumper wires in different electrical applications.

Connecting Wires:

Connecting wires allows an electrical current to travel from one point on a circuit to another because electricity needs a medium through which it can move. Most of the connecting wires are made up of copper or aluminum.

In this crystal ball project connecting wires are used to connect the circuit.

WORKING

How does it Work?

The voltage that the sensor outputs changes accordingly to the smoke/gas level that exists in the atmosphere. The sensor outputs a voltage that is proportional to the concentration of smoke/gas.

In other words, the relationship between voltage and gas concentration is the following:

- The greater the gas concentration, the greater the output voltage
- The lower the gas concentration, the lower the output voltage

The output can be an analog signal (A0) that can be read with an analog input of the Arduino or a digital output (D0) that can be read with a digital input of the Arduino.



Working Mechanism

Pin Wiring:

The MQ-2 sensor has 4 pins.

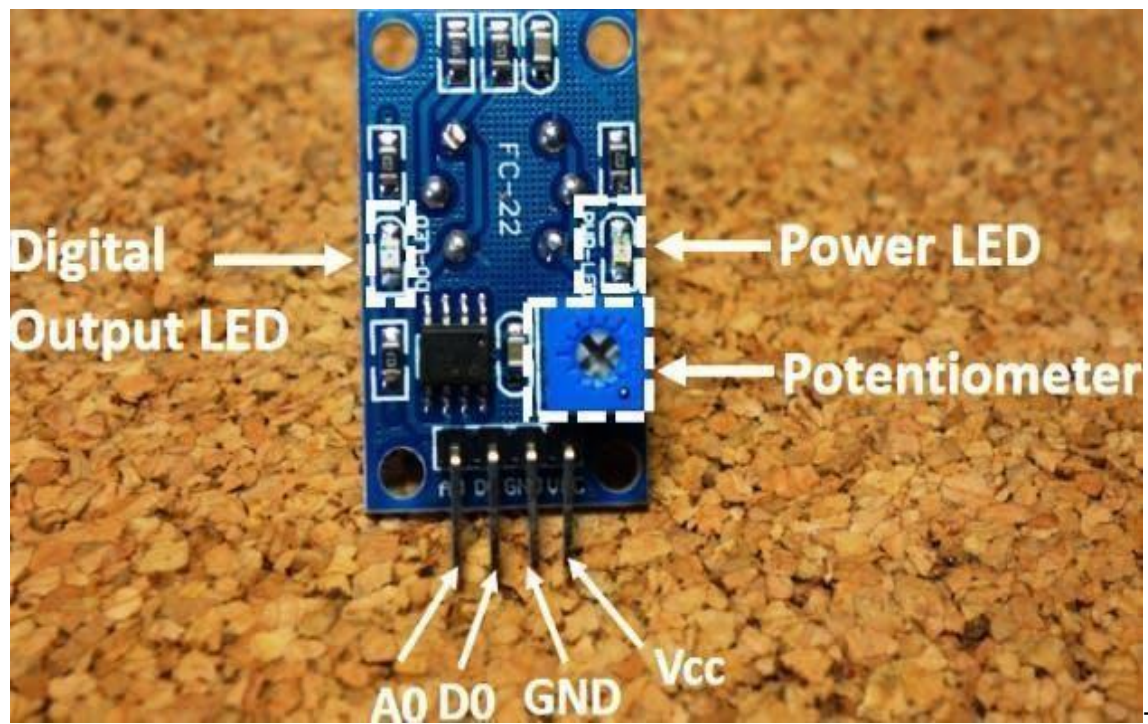
Pin-----Wiring to Arduino Uno

A0-----Analog pins

D0-----Digital pins

GND-----GND

VCC-----5V



CODE

```
#include<ESP8266WiFi.h> String apikey=
"GWAZAQD8TU9F7AFC". const
char*ssid="KL-UNIVERSITY4". const
char*pass="".
const char*server="api.thingspeak.com".
WiFiClient Client. void
setup ()
{
Serial.begin(9600). delay (10).
WiFi.begin(ssid,pass); while
(WiFi.status()!=WL_CONNECTED)
{delay (500).
Serial.print(".").
}
Serial.println("").
Serial.println("WiFi CONNECTED").
} void loop
()
{
int gas=analogRead(A0).
gas=map(gas,0,1023,0,100). if
(Client.connect(server,80))
{
String poststr=apikey;
poststr+="&field= ".
```



```
25 poststr+=String(gas).
   poststr="\r\n\r\n".
Client.print("post/update HTTP/1.1\n").
Client.print("Host: api.thingspeak.com\n").
Client.print("connection:close\n");
Client.print("x-THINGSPEAKAPIKEY:"+apikey+"\n").
Client.print("content-length:").
Client.print(poststr.length());
   Client.print("\n\n").
Client.print(poststr).
Serial.print("LPG GAS LEVEL IN %:").
Serial.println(gas).
}
Client.stop().
delay (500).
}
```

ADVANTAGES: -

A. User friendly: we can edit the code whenever we want to change tone of the button.

B. efficiency: it uses less power of only 5v.

C. portable: it is very small to carry wherever you go

D. Future Enhancement:

This technology could be further modified and more upgraded as per individual need and interest. We have discussed some basic ideas of this technology. And depending on innovative applications user can upgrade as per requirement.

- detection of low energy fires
- detection that is faster than heat detectors for most fires
- are preferred in life safety applications

OUTPUT: -

Hence the experiment of SMOKE DETECTOR is done by using MQ-2 smoke detecting sensor successfully and it is displayed.

CONCLUSION

This model can be used very easily instead of very large pianos and by installing a few more components we can make it very lighter and very easier to play and it will be very easier for the beginner in learning of piano.

- Smoke detector is one of the cheaper and easiest one.
- Most of industries use it, because it works fatly to protect, and it is most effective.
- Using this we can easily detect the smoke.
- LDS provide the location of leakages without any persons a different place.